

FOL Validator Component

Langflow custom component — validates and extracts FOL formulas before AST generation.

```
from langflow.custom import Component
from langflow.io import MessageTextInput, Output
from langflow.schema.message import Message
import json

class FOLValidator(Component):

    display_name = "FOL Validator"
    name = "FOLValidator"
    description = "Validates and extracts FOL formulas before AST generation"

    inputs = [
        MessageTextInput(
            name="input_value",
            display_name="Formalization JSON"
        )
    ]

    outputs = [
        Output(
            name="output",
            display_name="Validated FOL",
            method="build_output"
        )
    ]

    def build_output(self) -> Message:

        try:
            raw = (
                self.input_value.text
                if hasattr(self.input_value, "text")
                else str(self.input_value)
            )
            data = json.loads(raw)

        except Exception as e:
            return Message(
                text=json.dumps(
                    {
                        "fol_valid": False,
                        "error": str(e)
                    },
                    indent=2,
                    ensure_ascii=False
                )
            )

        errors = []

        for req in data.get(
```

```

        "requirements",
        []
    ):
        req_id = req.get(
            "requirement_id",
            "UNKNOWN"
        )
        fol = req.get(
            "fol_formula",
            ""
        )

        if not fol:
            errors.append(
                {
                    "requirement_id": req_id,
                    "error": "EMPTY_FOL_FORMULA"
                }
            )

    result = {
        "fol_valid":
            len(errors) == 0,
        "errors":
            errors,
        "requirements":
            data.get(
                "requirements",
                []
            )
    }

    return Message(
        text=json.dumps(
            result,
            indent=2,
            ensure_ascii=False
        )
    )

```